

Comments on errors made:

1. Many overly heated the given mixture (despite the question mentioned heat gently-bunsen flame should not be a strong one) as a result they obtained the black solid.
2. When describing the residue and filtrate, the physical state should be stated. For example, green solid and colourless solution and not green residue and colourless/white/clear filtrate.
3. In describing the observation of Cu^{2+} with aqueous ammonia, the student must state that light blue ppt is formed initially, before mentioning it dissolves in excess to produce a dark blue solution, instead of dark blue precipitate.
4. In explaining (f), students should explain which observation corresponds to the deduction of what cation or anion is present, and not simply putting down the observations only.
5. In (g), student needs to be careful in stating "cation is copper" and not copper is present. In fact, stating copper(II) or Cu^{2+} is the more straightforward way.
6. The standard tests for the presence of Cu^{2+} are using aqueous NaOH and aqueous NH_3 . As such, students should not use "forming a blue solution when an acid is added to the solid as evidence (since there could be other cations that may give rise to a blue solution-example vanadyl ion, VO^{2+} forms a blue solution in acid).

3. Washing soda are crystals of hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.

When exposed to the air, some of the water is lost from the crystals and a new substance is formed.

This process occurs faster in hotter climates. Plan an experiment to determine the percentage of water by mass present in the new substance.

You are provided with common laboratory apparatus.

[6]

Heating to dryness method max [6]:

M1 weigh (any) sample of washing soda to obtain mass of crystals using electronic balance

M2 heat (to remove water of crystallisation)

M3 in named container (must be: evaporating dish)

M4 reweigh & repeat heating

M5 to constant mass

M6 appropriate calculation suggested for the percentage of water mass of water

Comments on errors made:

1. Not stating the use of an electronic balance or simply mentioning weighing scale.
2. Instead of measuring the "mass" of students, they mentioned "weight".
3. Use a crucible/boiling/test-tube/beaker instead of an evaporating dish.
4. Did not mention reweighing and repeat heating to ensure constant mass.
5. Some thought the new substance formed was metal oxide, and carbon dioxide was given off.
6. Instead of taking the loss in mass of the solid over the mass of the solid, students stated over the mass of the "new" substance formed.

[Total: 6]